

Problem 73 (Putnam 2003). Show that for each positive integer n ,

$$n! = \prod_{i=1}^n \text{lcm}(1, 2, \dots, \lfloor n/i \rfloor).$$

Solution. Select an arbitrary prime p . We compute ν_p of both sides to show that they are equal. Notice that

$$\begin{aligned} \nu_p \left(\prod_{i=1}^n \text{lcm}(1, 2, \dots, \lfloor n/i \rfloor) \right) &= \sum_{i=1}^n \nu_p(\text{lcm}(1, 2, \dots, \lfloor n/i \rfloor)) \\ &= \sum_{i=1}^n \max_{1 \leq k \leq \lfloor n/i \rfloor} \nu_p(k) \\ &= \sum_{i=1}^n \lfloor \log_p(n/i) \rfloor \\ &= \sum_{i=1}^n \sum_{j \leq \log_p(n/i)} 1 \\ &= \sum_{j=1}^{\lfloor \log_p n \rfloor} \sum_{i \leq n/p^j} 1 \\ &= \sum_{j=1}^{\lfloor \log_p n \rfloor} \lfloor n/p^j \rfloor, \end{aligned}$$

so we are done by Legendre's formula. □

Problem 74 (Turkey TST 2005). Prove that for any positive integers $a_1, a_2, \dots, a_n$,

$$\prod_{1 \leq i < j \leq n} \frac{a_i - a_j}{i - j}$$

is an integer.

Solution. Fix an arbitrary prime p . For any set of numbers S , let $f(S, r, k)$ denote the number of elements in S which are congruent to r modulo p^k . Then

$$\nu_p \left(\prod_{\substack{x, y \in S \\ x \neq y}} (x - y) \right) = \sum_{\substack{x, y \in S \\ x \neq y}} \nu_p(x - y) = \sum_{k \geq 1} \sum_{r=0}^{p^k-1} \binom{f(S, r, k)}{2}$$

Let $n = |S| \geq p^k$. By Karamata's inequality, we have

$$\sum_{r=0}^{p^k-1} \binom{f(S, r, k)}{2} \geq (p^k - (n - p^k \lfloor n/p^k \rfloor)) \binom{\lfloor n/p^k \rfloor}{2} + (n - p^k \lfloor n/p^k \rfloor) \binom{\lfloor n/p^k \rfloor + 1}{2}.$$

Letting $A = \{a_1, a_2, \dots, a_n\}$ and $B = \{1, 2, \dots, n\}$, we see

$$\begin{aligned}
\nu_p\left(\prod_{1 \leq i < j \leq n} (a_i - a_j)\right) &= \sum_{k \geq 1} \sum_{r=0}^{p^k-1} \binom{f(A, r, k)}{2} \\
&\geq \sum_{k \geq 1} (p^k - (n - p^k \lfloor n/p^k \rfloor)) \binom{\lfloor n/p^k \rfloor}{2} + (n - p^k \lfloor n/p^k \rfloor) \binom{\lfloor n/p^k \rfloor + 1}{2} \\
&= \sum_{k \geq 1} \sum_{r=0}^{p^k-1} \binom{f(B, r, k)}{2} \\
&= \nu_p\left(\prod_{1 \leq i < j \leq n} (i - j)\right),
\end{aligned}$$

so we are done. □

Problem 75 (USAMO 2016). Prove that for any positive integer k ,

$$(k^2)! \times \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}$$

is an integer.

Solution. Using Legendre's formula, it suffices to show that

$$\left\lfloor \frac{k^2}{P} \right\rfloor \geq \sum_{j=0}^{k-1} \left(\left\lfloor \frac{j+k}{P} \right\rfloor - \left\lfloor \frac{j}{P} \right\rfloor \right)$$

for any prime power P . Since both sides are integral, it is enough to show that

$$\left\lfloor \frac{k^2}{P} \right\rfloor + 1 > \sum_{j=0}^{k-1} \left(\left\lfloor \frac{j+k}{P} \right\rfloor - \left\lfloor \frac{j}{P} \right\rfloor \right).$$

Writing $\lfloor x \rfloor = x - \{x\}$, it is equivalent to showing that

$$\left\{ \frac{k^2}{P} \right\} + \sum_{j=0}^{k-1} \left\{ \frac{j}{P} \right\} < 1 + \sum_{j=0}^{k-1} \left\{ \frac{j+k}{P} \right\}.$$

Let A denote the sum of the remainders when $0, 1, 2, \dots, k-1$ are divided by P , and B denote the sum of the remainders when $k, k+1, \dots, 2k-1$ are divided by P . We claim that $A \leq B$. This would imply that

$$\sum_{j=0}^{k-1} \left\{ \frac{j}{P} \right\} \leq \sum_{j=0}^{k-1} \left\{ \frac{j+k}{P} \right\}$$

Trivially, $\{k^2/P\} < 1$, so this is enough to solve the problem.

We now resolve the claim. Let r be the remainder of k modulo P . If $2r \geq P$, then

$$B = r + (r+1) + \dots + (P-1) + A - ((r-1) + (r-2) + \dots + (r-(P-r))).$$

Since $r+i > r-i-1$, we have $B \geq A$.

On the other hand, if $2r < P$, then

$$B = r + (r+1) + \dots + (2r-1) + A - (0+1+\dots+(r-1)),$$

which is obviously larger than A . □

Problem 76 (Generalization of Romania TST 2007). Find all positive integers (x, y, n) such that

$$x^n - y^n = x! - y!.$$

Solution. Obviously (x, x, n) is a solution for any x and n . The only other solutions are $(2, 1, 1)$, $(1, 2, 1)$ which can easily be seen to work. Without loss of generality assume $x > y$.

When $n = 1$, we have $x! - y! = y!(x(x-1) \cdots (y+1) - 1) \geq x-1 \geq x-y$. Equality holds when $(x, y) = (2, 1)$.

We show that there are no other solutions when $n > 1$. First we show that $x \leq 2n$. Suppose on the contrary that $x > 2n \geq 4$. Since $y < x$,

$$x! - y! \geq (x-1)!(x-1) = (x-1)^2[2(x-2)][3(x-3)] \cdots$$

Since $k(x-k) > x$ for $k \geq 2$ and $x > 4$, it follows that

$$x! - y! > (x-1)^2 x^{\frac{x-2}{2}} > x^n > x^n - y^n,$$

since $x > 2n$, contradiction.

Therefore $y < x \leq 2n$. Trivially, $y > 1$. Pick a prime divisor p of y . Since p divides $x! - y!$, it must also divide x . Hence $\nu_p(x^n - y^n) \geq n$. On the other hand,

$$\nu_p(x! - y!) = \nu_p(y!) = \sum_{k \geq 1} \left\lfloor \frac{y}{p^k} \right\rfloor < \frac{y}{p-1}.$$

We have shown that $n < \frac{y}{p-1} < \frac{2n}{p-1}$. The only possibility is $p = 2$ and $y > n$. This means y is a power of 2. We select an odd prime divisor q of x . (Note that x cannot be a power of 2 or else $x \geq 2y > 2n$.) Then q does not divide $x^n - y^n$, so $q \nmid y!$, implying that $q > y$. Therefore $x = q$, or else $x \geq 2q > 2y > 2n$. However, $x! - y! \not\equiv x^n - y^n \pmod{2}$, so we are done. $\square$